

Title Page

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Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch

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The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement:

The source code, simulation scripts, benchmark results, plotting scripts, derived plant-historian metrics, and bounded controller-export excerpts are available at <https://github.com/VickylastShao/Robust-HOCBF-Safety-Filtering-for-Supercritical-Power-Plants-under-Model-Mismatch>.

Complete raw plant records are proprietary enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and an executed data-use agreement.

Author Contributions:

Zhuang Shao conceived the study, developed the methodology, performed the simulations, analyzed the results, and wrote the original draft.

Lijun Lei provided engineering requirements, plant-data interpretation, and field-validation support.

Peng Wang contributed to control-system analysis and manuscript review.

Liang Zheng supported plant-data preparation and validation-log interpretation.

Jie Zhou contributed to deployment-context analysis and manuscript review.

All authors reviewed and approved the final manuscript.

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